

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Game Based Learning Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 40 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A Game Based Learning Assessment

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Understanding of the Problem / Game Objective	10%	
Application of AI Search / Problem-Solving Technique	15%	
Successful Completion of the Game / Puzzle	20%	
Efficiency of Solution / Optimal Moves	10%	
Documentation / Evidence of Completion	15%	
Understanding of the Problem / Game Objective	15%	
Originality and Critical Thinking	15%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

Answer All the Questions

Q. No.	Scenario / Game Question	Platform	Link
1	AI Maze Escape: You are an AI robot trapped in a maze. Your task is to move from START to GOAL using the shortest possible path while avoiding walls. Challenge: Find the shortest path and record the number of steps taken.	Scratch	https://scratch.mit.edu
2	N-Queens Challenge: You are an AI chess strategist. Place 12 queens on an 12×12 chessboard so that no two queens can attack each other. Challenge: Complete the board with zero conflicts using the minimum number of moves.	Online N-Queens Game	https://www.n-queens.com
3	Water Jug Puzzle: You are an AI agent solving a water jug problem. You have two jugs with capacities of 11 litres and 9 litres. Neither jug has measurement markings. Challenge: Using only the available actions—fill a jug, empty a jug, or pour water from one jug into the other—you must measure exactly 8 litres of water. Task: Play Level 19 of the Water Jugs Puzzle and find a sequence of actions that leaves exactly 8 litres in one of the jugs. Try to solve the puzzle using the minimum number of moves.	Tic-Tac-Toe	https://playtictactoe.org
4	Connect Four AI Challenge: You are playing against an AI opponent. Your goal is to connect four of your pieces in a row before the AI does. Challenge: Plan your moves carefully and defeat the AI opponent.	Connect Four	https://www.mathsisfun.com/games/connect4.html

5	8-Puzzle AI Challenge: You are given a scrambled 8-puzzle. Move the tiles one at a time and arrange them in the correct order. Challenge: Solve the puzzle using the minimum possible number of moves.	8-Puzzle Game	https://8puzzle.org/
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Structure of the Report – Game-Based Learning

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of the Game-Based Activity	Provide the name of the game or puzzle and the level/challenge attempted.
2	Game Platform / Link	Mention the platform or website used and provide the corresponding game link.
3	Objective of the Activity	Explain the objective of the game and the target that needs to be achieved.
4	Problem Statement	Describe the problem or challenge given in the game. What is the student required to solve?
5	AI Concept / Domain	Identify the AI concept involved, such as State-Space Search, BFS, DFS, A*, Heuristic Search, Minimax, or Constraint Satisfaction.
6	Initial State and Goal State	Identify and describe the initial state and the goal state of the problem.
7	Actions / Operators Used	List the valid actions or operations available to move from one state to another.
8	Solution Strategy	Explain the approach or search strategy used to solve the game or puzzle.
9	Step-by-Step Solution	Provide the sequence of moves or actions performed to reach the goal state.
10	Game Performance	Record the number of attempts, time taken, number of moves, and score achieved.
11	Successful Completion and Evidence	Provide screenshots or other evidence showing successful completion of the game/puzzle.
12	Efficiency and Optimality	Analyse whether the solution was optimal or efficient. Can the puzzle be solved using fewer moves or less time?
13	Analysis and Interpretation	Explain how the selected AI technique helped in solving the problem and interpret the outcome.
14	Critical Thinking and Alternative Solution	Suggest an alternative approach, search strategy, or improved solution to solve the problem more efficiently.
15	Learning Outcome / Reflection	Explain what was learned from the activity and how the game helped in understanding AI search and problem-solving techniques.
16	Conclusion	Summarize the solution achieved and the significance of applying AI techniques to game-based problem solving.
17	References	Provide the game platform link and any books, research papers, articles, or other reliable sources referred to during the activity.

Evaluation Rubrics

Criteria	Percentage (%)	Excellent	Good	Satisfactory	Needs Improvement
Understanding of the Problem / Game Objective	10%	Clearly understands the problem and game objective	Understands the objective with minor errors	Partially understands the problem	Does not understand the objective
Application of AI Search / Problem-Solving Technique	15%	Correctly applies the appropriate AI search or problem-solving technique	Applies the technique with minor errors	Shows limited application of the technique	Unable to apply the appropriate technique
Successful Completion of the Game / Puzzle	20%	Successfully solves/completes the game independently	Completes the game with minimal assistance	Partially completes the challenge	Unable to complete the challenge
Efficiency of Solution / Optimal Moves	10%	Solves using optimal or highly efficient moves	Uses a reasonably efficient sequence of moves	Uses several unnecessary moves	Solution is highly inefficient or incomplete
Documentation / Evidence of Completion	15%	Provides clear steps, score, screenshots, and evidence of completion	Provides most required evidence	Provides incomplete steps or evidence	Provides no proper documentation or evidence
Analysis and Interpretation of the Solution	15%	Clearly analyses the solution and explains the AI technique and outcome	Provides a good analysis with minor gaps	Provides limited analysis or explanation	Unable to analyse or interpret the solution
Originality and Critical Thinking	15%	Demonstrates excellent independent thinking and explores an effective or alternative solution	Demonstrates good independent thinking	Shows limited critical thinking	Shows little or no independent thinking